Hyperbolic Functions

Questions 1–231

2.14 — Question 1. By considering Osborn's Rule, or otherwise, write down the analogous hyperbolic identity for each of these trigonometric identities:

a) $\sin^2 x + \cos^2 x \equiv 1$
b) $1 + \cot^2 x \equiv \operatorname{cosec}^2 x$
c) $\tan^2 x + 1 \equiv \sec^2 x$
d) $\sin(x + y) \equiv \sin x \cos y + \cos x \sin y$
e) $\sin(x - y) \equiv \sin x \cos y - \cos x \sin y$
f) $\cos(x + y) \equiv \cos x \cos y - \sin x \sin y$
g) $\cos(x - y) \equiv \cos x \cos y + \sin x \sin y$
h) $\tan(x + y) \equiv \dfrac{\tan x + \tan y}{1 - \tan x \tan y}$
i) $\tan(x - y) \equiv \dfrac{\tan x - \tan y}{1 + \tan x \tan y}$
j) $\sin 2x \equiv 2 \sin x \cos x$
k) $\cos 2x \equiv \cos^2 x - \sin^2 x$
l) $\cos 2x \equiv 2 \cos^2 x - 1$
m) $\cos 2x \equiv 1 - 2 \sin^2 x$
n) $\tan 2x \equiv \dfrac{2 \tan x}{1 - \tan^2 x}$
o) $\sin 3x \equiv 3 \sin x - 4 \sin^3 x$
p) $\cos 3x \equiv 4 \cos^3 x - 3 \cos x$
q) $\tan 3x \equiv \dfrac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$
r) $\sin 4x \equiv \cos x(4 \sin x - 8 \sin^3 x)$
s) $\cos 4x \equiv 8 \cos^4 x - 8 \cos^2 x + 1$
t) $\tan 4x \equiv \dfrac{4 \tan x - 4 \tan^3 x}{1 - 6 \tan^2 x + \tan^4 x}$

2.14 — Question 2. Write the following hyperbolic expressions in terms of $\tanh x$, $\operatorname{cosech} x$, $\operatorname{sech} x$ and $\coth x$ only:

a) $\dfrac{\sinh x}{\cosh^2 x}$
b) $\dfrac{\cosh x}{\sinh^2 x}$

2.14 — Question 3. Simplify the following hyperbolic expressions as far as possible:

a) $\operatorname{cosech} x \sinh x \cosh x$

b) $\tanh x \operatorname{sech} x \cosh x$

c) $\tanh x \operatorname{cosech} x \coth x \sinh x$

2.14 — Question 4. Simplify the following hyperbolic expressions, removing any quotients where possible:

a) $\dfrac{\operatorname{sech} x}{\operatorname{cosech} x}$

b) $\dfrac{\coth x}{\operatorname{sech} x}$

c) $\dfrac{\operatorname{cosech} x}{\tanh x}$

d) $\dfrac{\operatorname{sech} x}{\coth x}$

e) $\dfrac{\operatorname{sech}^2 x}{\coth x}$

f) $\dfrac{\operatorname{cosech}^2 x \coth^2 x}{\operatorname{sech} x \sinh x}$

2.14 — Question 5. Given that k is a constant where $k > 1$, show that

$$-\ln(k + \sqrt{k^2 - 1}) = \ln(k - \sqrt{k^2 - 1}).$$

2.14 — Question 6. Given that k is a constant, show that

$$-\ln(k + \sqrt{k^2 + 1}) \neq \ln(k - \sqrt{k^2 + 1}).$$

2.14 — Question 7. Show that the equation $\cosh x = k$, where k is a constant such that $k > 1$, has two real solutions: $x = \ln(k + \sqrt{k^2 - 1})$ and $x = \ln(k - \sqrt{k^2 - 1})$.

2.14 — Question 8. It is given that $\cosh x = 4$. Show that $x = \ln(4 \pm \sqrt{15})$ and hence $x = \pm \ln(4 + \sqrt{15})$.

2.14 — Question 9. Express each of these expressions in terms of $\cosh x$:

a) $8\cosh^2 x + 7\sinh^2 x$

b) $3\sinh^2 x - 13\cosh^2 x$

2.14 — Question 10. Solve $3\cosh^2 x - \sinh^2 x = 5$, writing your answers in exact form.

2.14 — Question 11. Solve each of these equations, writing your answers in exact form:

a) $\cosh^2 x - \sinh x - 7 = 0$
b) $\sinh^2 x - 2\cosh x - 7 = 0$
c) $\cosh^2 x - 14\sinh x + 47 = 0$
d) $2\sinh^2 x - 13\cosh x + 8 = 0$
e) $3\cosh^2 x + 22\sinh x - 19 = 0$
f) $4\sinh^2 x - 4\cosh x - 11 = 0$

2.14 — Question 12. Solve each of these equations, writing your answers in exact form:

a) $25\cosh^4 x - 40\cosh^2 x - 9 = 0$
b) $36\cosh^4 x - 36\sinh^2 x - 91 = 0$
c) $192\sinh^4 x - 172\cosh^2 x + 67 = 0$

2.14 — Question 13. Starting from $\cosh^2 x - \sinh^2 x \equiv 1$:

a) derive an identity for $\coth x$ and $\operatorname{cosech} x$

b) derive an identity for $\tanh x$ and $\operatorname{sech} x$

2.14 — Question 14. Solve each of these equations, writing your answers in exact form:

a) $2\operatorname{sech}^2 x - 3\tanh x = 0$

b) $3\tanh^2 x - 5\operatorname{sech} x - 1 = 0$

c) $\operatorname{cosech}^2 x - 2\coth x - 7 = 0$

d) $4\coth^2 x - 25\operatorname{cosech} x + 2 = 0$

2.14 — Question 15. Solve each of these equations, writing your answers in exact form:

a) $8\operatorname{sech}^2 x - 6\tanh x + 1 = 0$

b) $10\tanh^2 x - 21\operatorname{sech} x + 3 = 0$

c) $24\operatorname{cosech}^2 x + 58\coth x + 59 = 0$

d) $25\coth^2 x - 73\operatorname{cosech} x = 31$

2.14 — Question 16. Prove there are no real solutions to the equation $6\tanh^2 x + 17\operatorname{sech} x - 18 = 0$.

2.14 — Question 17. Solve each of these equations, writing your answers in exact form:

a) $\sinh x = \operatorname{sech} x$

b) $\tanh x = \operatorname{sech} x$

c) $\sinh x = 2\coth x$

d) $3\cosh x = \coth x$

2.14 — Question 18.

a) Sketch the graphs of $y = 2\cosh x$ and $y = \operatorname{cosech} x$ on the same axes.

b) Determine the exact coordinates of where the two graphs intersect.

2.14 — Question 19. It is given that $\sinh x = 3$. Find the exact values of:

a) $\cosh x$
b) $\tanh x$
c) $\coth x$
d) $\operatorname{cosech} x$
e) $\operatorname{sech} x$

2.14 — Question 20. It is given that $\operatorname{sech} x = \dfrac{1}{\sqrt{2}}$. Find the exact values of:

a) $\tanh x$
b) $\cosh x$
c) $\sinh x$
d) $\operatorname{cosech} x$
e) $\coth x$

2.14 — Question 21. It is given that $48\coth^2 x + 34\,\text{cosech}\, x - 53 = 0$ and x is positive. Find the exact values of:

a) $\text{cosech}\, x$
b) $\sinh x$
c) $\cosh x$
d) $\text{sech}\, x$
e) $\tanh x$
f) $\coth x$

2.14 — Question 22. Simplify the following hyperbolic expressions:

a) $\dfrac{\text{sech}^2 x}{1 - \tanh^2 x}$
b) $\dfrac{\coth^2 x - 1}{\text{cosech}^2 x}$
c) $\dfrac{1 - \tanh^2 x}{\coth^2 x - 1}$
d) $\dfrac{3 - 3\coth^2 x}{8 - 8\tanh^2 x}$

2.14 — Question 23. It is given that $\sinh(1+\ln 6) = \dfrac{36e^2-1}{12e}$.

a) Hence find the exact value of $\cosh(1+\ln 6)$ in terms of e.

b) Using your answer to (a), show that $\tanh(1+\sqrt{6}) = \dfrac{36e^2-1}{36e^2+1}$.

2.14 — Question 24. It is given that $p(x) = 12x^3 - 7x^2 - 42x + 40$.

a) Show that $(4x-5)$ is a factor of $p(x)$.

b) Write $p(x)$ as a product of three linear factors.

c) Hence prove that there are no real solutions to the equation

$$12\operatorname{sech}^3\theta + 7\tanh^2\theta - 42\operatorname{sech}\theta + 33 = 0.$$

2.14 — Question 25.

a) Find $\sinh(\ln 3)$ and $\cosh(\ln 3)$.

b) Find $\sinh(\ln 4)$ and $\cosh(\ln 4)$.

c) Using your answers to (a) and (b), find the exact value of $\sinh(\ln 4 - \ln 3)$.

2.14 — Question 26.

a) Find $\sinh(\ln 5)$ and $\cosh(\ln 5)$.

b) Find $\sinh(\ln 6)$ and $\cosh(\ln 6)$.

c) Using your answers to (a) and (b), find the exact value of $\cosh(\ln 6 - \ln 5)$.

2.14 — Question 27.

a) Find $\tanh(\ln 2)$.

b) Find $\tanh(\ln 7)$.

c) Using your answers to (a) and (b), find the exact value of $\tanh(\ln 2 + \ln 7)$.

2.14 — Question 28. Write the following in the form $\sinh k$, where k is a constant:

a) $\sinh 3 \cosh 6 - \cosh 3 \sinh 6$

b) $\sinh 4 \cosh 5 + \cosh 4 \sinh 5$

2.14 — Question 29. Write the following in the form $\cosh k$, where k is a constant:

a) $\cosh 3 \cosh 7 - \sinh 3 \sinh 7$

b) $\cosh 8 \cosh 9 + \sinh 8 \sinh 9$

2.14 — Question 30. It is given that $\cosh x = 2$ and $\cosh y = 3$, where x and y are real. Find the exact value of:

a) $\cosh(x - y)$

b) $\sinh(x - y)$

2.14 — Question 31. Simplify the following expressions:

a) $\sinh(x + \ln 2) + \cosh(x - \ln 2)$

b) $\sinh(x - \ln 3) - \cosh(x + \ln 3)$

c) $\cosh(x - \ln 4) - \sinh(x - \ln 4)$

2.14 — Question 32. In each case, find the exact value that solves the equation:

a) $\sinh(x + \ln 2) = \cosh x$

b) $\cosh(x - \ln 3) = \cosh x$

c) $\cosh(x - \ln 4) = \sinh x$

2.14 — Question 33. In each case, find the exact value that solves the equation:

a) $\sinh(x + \ln 2) = \cosh(x - \ln 4)$

b) $\cosh(x - \ln 9) = \sinh(x + \ln 3)$

c) $\cosh(x + 3\ln 2) = \cosh(x - \ln 20)$

2.14 — Question 34. Prove the following hyperbolic identities:

a) $\coth x + \cosh x \equiv \coth x(1 + \sinh x)$

b) $\coth x + \operatorname{sech} x \equiv \operatorname{cosech} x \operatorname{sech} x(\sinh x + \cosh^2 x)$

c) $\coth x - \tanh x \equiv \operatorname{cosech} x \operatorname{sech} x$

2.14 — Question 35. Prove the following hyperbolic identities:

a) $\cosh x - \operatorname{sech} x \equiv \sinh x \tanh x$

b) $\operatorname{cosech} x + \sinh x \equiv \cosh x \coth x$

c) $\operatorname{cosech} x \operatorname{sech} x \equiv \tanh x(\coth^2 x - 1)$

d) $\dfrac{\operatorname{cosech} x(\sinh x \tanh x - 1)}{\sinh x + \cosh x} \equiv \tanh x - \operatorname{cosech} x$

2.14 — Question 36. Simplify the following expressions:

a) $8 \sinh 2x \cosh 2x$

b) $2 \cosh^2 5x - 1$

c) $1 + 2 \sinh^2 3x$

d) $9 \cosh^2 \frac{x}{4} + 9 \sinh^2 \frac{x}{4}$

2.14 — Question 37. Solve the following equations, giving your answers in an exact form:

a) $\sinh 2\theta - 2\sinh\theta = 0$

b) $\sinh 2\theta = 3\sinh\theta$

c) $5\sinh 2\theta = 6\cosh\theta$

d) $8\cosh\theta - 3\sinh 2\theta = 0$

2.14 — Question 38. Solve the following equations, giving your answers in an exact form:

a) $2\cosh 2\theta = 7\cosh\theta$

b) $3\cosh 2\theta + 11\sinh\theta = 0$

c) $3\sinh\theta - \cosh 2\theta = 0$

d) $\cosh 2\theta - 4\cosh\theta = 0$

2.14 — Question 39. Solve the following equations, giving your answers in an exact form:

a) $\cosh 2\theta + 2\sinh\theta - 5 = 0$

b) $5\cosh 2\theta - 3\sinh\theta = 5$

c) $8\cosh\theta - 3\cosh 2\theta = 3$

d) $5\cosh\theta - 12\cosh 2\theta + 17 = 0$

2.14 — Question 40. Solve the following equations, giving your answers in an exact form:

a) $2\sinh 2\theta\,\operatorname{cosech}\theta = 5$

b) $5\sinh 2\theta\,\operatorname{sech}\theta = 7$

c) $\cosh 2\theta\,\operatorname{cosech}\theta = 3$

d) $2\cosh 2\theta\,\operatorname{sech}\theta = 7$

2.14 — Question 41. Solve the following equations, giving your answers in an exact form:

a) $5\tanh 2x - 9\tanh x = 0$

b) $4\tanh x - 3\tanh 2x = 0$

2.14 — Question 42. It is given that $\sinh x = \frac{4}{3}$. Find the exact value of $\sinh 2x$.

2.14 — Question 43. It is given that $\sinh x = -\frac{1}{3}$. Find the exact value of $\cosh 2x$.

2.14 — Question 44. It is given that $\cosh x = \frac{9}{4}$. Find the exact value of $\tanh 2x$.

2.14 — Question 45.

a) Use hyperbolic identities, including $\sinh(A \pm B) \equiv \sinh A \cosh B \pm \cosh A \sinh B$, to write $\sinh 3\theta$ in terms of $\sinh \theta$.

It is given that $x = \sinh \theta$ and that $20x^3 + 15x - 12 = 0$.

b) Find the value of θ in the form $p \ln q$ where p is a rational number and q is a positive integer.

c) Hence find the real root of the equation $20x^3 + 15x - 12 = 0$, giving your answer in the form $\frac{1}{2}(5^m - 5^n)$, where m and n are rational numbers.

2.14 — Question 46. Prove the following hyperbolic identities:

a) $\coth^2 x - \tanh^2 x \equiv \operatorname{cosech}^2 x + \operatorname{sech}^2 x$

b) $\sinh^2 x \tanh^2 x \equiv \sinh^2 x - \tanh^2 x$

c) $\operatorname{cosech}^2 x - \operatorname{sech}^2 x \equiv 4 \operatorname{cosech}^2 2x$

2.14 — Question 47. Prove the following hyperbolic identities:

a) $(\operatorname{sech} x - \tanh x)^2 \equiv 1 - 2\operatorname{sech} x \tanh x$

b) $(\operatorname{sech} x - \coth x)^2 \equiv 2 - \tanh^2 x - 2\operatorname{cosech} x + \operatorname{cosech}^2 x$

c) $(\operatorname{sech} x - \operatorname{cosech} x)^2 \equiv \operatorname{cosech}^2 x \operatorname{sech}^2 x(\cosh 2x - \sinh 2x)$

2.14 — Question 48.

a) Prove the following hyperbolic identity:

$$\frac{1+\cosh x}{\sinh x} - \frac{\sinh x}{1+\cosh x} \equiv 2\operatorname{cosech} x.$$

b) Hence, where $t \neq 0$, solve the equation

$$\frac{1+\cosh t}{\sinh t} - \frac{\sinh t}{1+\cosh t} = -4.$$

2.14 — Question 49.

a) Prove the following hyperbolic identity:

$$\frac{\sinh x}{\cosh x - 1} - \operatorname{cosech} x \equiv \coth x.$$

b) Hence, given $t \neq 0$, solve the equation

$$\frac{\sinh t}{\cosh t - 1} = 8 + \coth t.$$

2.14 — Question 50. Show that the solution to the equation $\operatorname{sech} \frac{x}{2} = 6 \sinh \frac{x}{2}$ can be written in the form $\ln(p + \sqrt{q}) - \ln r$, where p, q and r are positive integers.

2.14 — Question 51. Prove the following hyperbolic identities:

a) $\coth 2x - \tanh 2x \equiv 2\,\text{cosech}\, 4x$

b) $\coth x - \tanh 2x \equiv \coth x\,\text{sech}\, 2x$

2.14 — Question 52. Prove the following hyperbolic identities:

a) $\coth 2x - \cosh 2x \equiv \coth 2x(1 - \sinh 2x)$

b) $\coth 2x + \text{cosech}\, 2x \equiv \coth x$

2.14 — Question 53. Prove the following hyperbolic identity:

$$2\operatorname{cosech} 4x(\sinh^2 x + \cosh 2x) \equiv \tanh 2x + \operatorname{cosech} 2x.$$

2.14 — Question 54. Prove the following hyperbolic identities:

a) $(\sinh 2x + \cosh 2x)^2 \equiv \sinh 4x + \cosh 4x$

b) $\dfrac{2\sinh x}{\cosh 2x - 1} - \dfrac{2\sinh x}{\cosh 2x + 1} \equiv \operatorname{cosech} x \operatorname{sech}^2 x$

2.14 — Question 55. Prove the following hyperbolic identities:

a) $\cosh 2x - \sinh 2x \equiv \dfrac{1}{(\sinh x + \cosh x)^2}$

b) $\coth x(\cosh 2x - \sinh 2x) \equiv \dfrac{\cosh x(\coth x - 1)}{\sinh x + \cosh x}$

2.14 — Question 56. Solve the equation $8\sinh x + 15\cosh x = 6\sinh x \cosh x + 20$, giving your answer in exact form.

2.14 — Question 57. Solve the following equations:

a) $\sinh(\ln(8+2x)) = x$

b) $\cosh(\ln(3-5x)) + 10x = 0$

c) $\operatorname{sech}(\ln(4x+9)) + \frac{1}{12x} = 0$

d) $\tanh(\ln(\sqrt{3}+4x)) = 2x$

2.15 — Question 1. For each of the following functions, sketch the graph and state whether it is one-to-one or if it is many-to-one.

a) $y = \sinh x$

b) $y = \cosh x$

c) $y = \tanh x$

The answers for the following questions which require a range of values should be given in set notation.

2.15 — Question 2. State the largest possible domain and range of the following functions:

a) $y = \sinh x$

b) $y = \cosh x$

c) $y = \tanh x$

2.15 — Question 3. State the range of the following, given $x < 0$:

a) $y = \sinh x$

b) $y = \cosh x$

c) $y = \tanh x$

2.15 — Question 4. State the range of the following, given $x \in \mathbb{R}$:

a) $f(x) = \sinh^2 x$
b) $y = \cosh x + 3$
c) $g(x) = \tanh(x - 2)$
d) $y = 2\sinh x$
e) $h(x) = -\cosh^2 x$
f) $y = \tanh^2 x + 5$

2.15 — Question 5.

a) Write $2\sinh x + 4\cosh x$ in the form $ae^x + be^{-x}$ where a and b are real constants.
b) Hence explain why, for $x \in \mathbb{R}$, that $2\sinh x + 4\cosh x > e^{-x}$.

2.15 — Question 6. Show clearly that, for $x \in \mathbb{R}$,

$$4\sinh^2 x - e^{-2x} > -2.$$

Prerequisite knowledge for Q7: Hyperbolic Double-Angle Formulae

2.15 — Question 7.

a) Write $\sinh 2x$ in terms of $\sinh x$ and $\cosh x$.

b) Given $x > 0$, write down the range of:

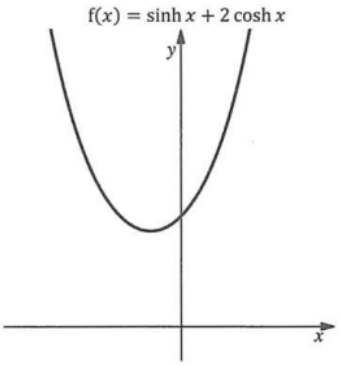
i) $y = \sinh x$

ii) $y = \cosh x$

c) Hence show that $\sinh 2x > \sinh x$ for $x > 0$.

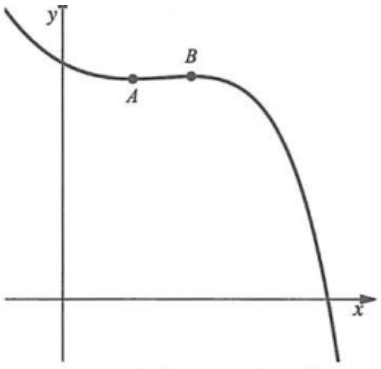
Prerequisite knowledge for Q8–Q10: Differentiation of Hyperbolic Functions

2.15 — Question 8. The graph below shows $f(x) = \sinh x + 2\cosh x$.



Find the exact largest possible range of f, writing your answer in set notation.

2.15 — Question 9. The graph below shows the curve with equation $y = 3\cosh x - \sinh x \cosh x$. The points A and B on the curve are stationary points and have x-coordinates a and b respectively.



It is given that the function f is defined as $f(x) = 3\cosh x - \sinh x \cosh x$ where $x \in [a, b]$. Find the range of f in interval notation.

2.15 — Question 10. It is given that

$$f(x) = \frac{e^{2x} + e^{-2x}}{(e^x + e^{-x})^2} - \tanh^2 x, \qquad x \in \mathbb{R}.$$

a) Show that $f(x) = 2(e^x + e^{-x})^{-2}$, $x \in \mathbb{R}$.
b) Find the coordinates of the stationary point of f.
c) State the range of f, giving your answer in set notation.

2.16 — Question 1. For each of the following functions, sketch the graph:
a) $y = \operatorname{arsinh} x$
b) $y = \operatorname{arcosh} x$
c) $y = \operatorname{artanh} x$

2.16 — Question 2. Given that $x \in \mathbb{R}$, show that

$$\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2+1}\right).$$

2.16 — Question 3. Given that $x \in \mathbb{R}$ and $k > 0$, show that

$$\operatorname{arsinh} \frac{x}{k} = \ln\left(\frac{x + \sqrt{x^2+k^2}}{k}\right).$$

2.16 — Question 4. Given that $x \geq 1$, show that

$$\operatorname{arcosh} x = \ln\left(x + \sqrt{x^2 - 1}\right).$$

2.16 — Question 5. Given that $\frac{x}{k} \geq 1$ and $k > 0$, show that

$$\operatorname{arcosh} \frac{x}{k} = \ln\left(\frac{x + \sqrt{x^2 - k^2}}{k}\right).$$

2.16 — Question 6. Given that $-1 < x < 1$, show that

$$\operatorname{artanh} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right).$$

2.16 — Question 7. Given that $-k < x < k$ and $k > 0$, show that

$$\operatorname{artanh} \frac{x}{k} = \frac{1}{2} \ln \left(\frac{k+x}{k-x} \right).$$

2.16 — Question 8. Solve the equation $\operatorname{arcosh} x = \ln 10$, writing your answer in exact form.

2.16 — Question 9. Solve the equation $\operatorname{arsinh} x = -\ln 6$, writing your answer in exact form.

2.16 — Question 10. Solve the equation $\operatorname{artanh} \frac{x}{6} = -\ln 2$, writing your answer in exact form.

2.16 — Question 11. Solve the following equations, writing your answers in exact form:

a) $\operatorname{artanh} x = \ln\left(\sqrt{x+2}\right)$

b) $\operatorname{artanh} \frac{x}{3} = \ln\left(\sqrt{3-2x}\right)$

c) $\operatorname{artanh} 5x = \ln\left(\sqrt{2+5x}\right)$

d) $4x + 1 = \tanh(\ln(\sqrt{5-2x}))$

2.16 — Question 12. State the largest possible domain and range of the following functions:

a) $y = \operatorname{arsinh} x$

b) $y = \operatorname{arcosh} x$

c) $y = \operatorname{artanh} x$

2.16 — Question 13. State the exact range of the following functions, given $-\frac{1}{2} < x < \frac{1}{2}$:

a) $f(x) = \operatorname{arsinh} x$

b) $g(x) = \operatorname{artanh} x$

2.16 — Question 14. State the largest possible domain and range of the following functions:

a) $y = \operatorname{arcosh} x + 1$
b) $y = \operatorname{arcosh}(x - 1)$
c) $y = \operatorname{arcosh} 2x$
d) $y = 2 \operatorname{arcosh} x$
e) $y = 3 \operatorname{arcosh}(x + 2)$
f) $y = \operatorname{arcosh}(2x - 3)$
g) $y = 4 \operatorname{arcosh} x - 5$
h) $y = \operatorname{arcosh}(4 - x)$
i) $y = 8 - 3 \operatorname{arcosh} x$
j) $y = -\operatorname{arcosh}\left(9 - \frac{1}{2}x\right)$

2.16 — Question 15. State the largest possible domain and range of the following functions:

a) $y = \operatorname{arsinh}^2 x$
b) $y = \operatorname{arcosh}^2 x$
c) $y = \operatorname{artanh}^2 x$

2.16 — Question 16. Given that $x \geq 1$, explain why $\operatorname{arsinh} x > \operatorname{arcosh} x$.

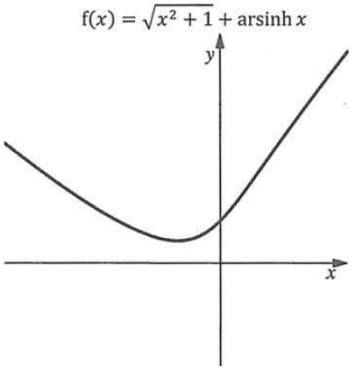
2.16 — Question 17. Given that $x \geq 1$, state the exact range of $f(x) = \operatorname{arsinh} x + \operatorname{arcosh} x$.

2.16 — Question 18. Solve the equation

$$\operatorname{artanh} x + \ln\sqrt{1-x} = \frac{1}{2}\ln 3 - 2\ln 2.$$

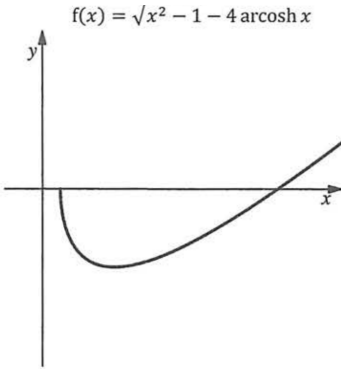
2.16 — Question 19. Solve the equation $\operatorname{arsinh} x = \ln 4 - \ln 3 + \operatorname{arcosh} x$.

2.16 — Question 20. The graph below shows $f(x) = \sqrt{x^2+1} + \operatorname{arsinh} x$.



Find the exact largest possible range of f, writing your answer in set notation.

2.16 — Question 21. The graph below shows $f(x) = \sqrt{x^2-1} - 4\operatorname{arcosh} x$.



Find the exact largest possible range of f, writing your answer in set notation.

2.16 — Question 22. It is given that $f(x) = 2\operatorname{arsinh} x - \operatorname{arcosh} x$.

a) Explain why the largest possible domain of f is $\{x : x \in \mathbb{R},\ x \geq 1\}$.

b) Show that $y = f(x)$ has a single stationary point.

c) Show that the range of f is greater than or equal to

$$2\ln(\sqrt{15} + 2\sqrt{6}) - \ln(\sqrt{15} + \sqrt{6}) - \ln 3.$$

2.16 — Question 23. Given that $x \neq 0$, show that

$$\operatorname{arcosech} x = \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} + 1}\right).$$

2.16 — Question 24. Given that $x \neq 0$ and $k > 0$, show that

$$\operatorname{arcosech} \frac{x}{k} = \ln\left(\frac{k + \sqrt{k^2 + x^2}}{x}\right).$$

2.16 — Question 25. Given that $0 < x < 1$, show that

$$\operatorname{arsech} x = \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} - 1}\right).$$

2.16 — Question 26. Given that $0 < \frac{x}{k} < 1$ and $k > 0$, show that

$$\operatorname{arsech} \frac{x}{k} = \ln\left(\frac{k + \sqrt{k^2 - x^2}}{x}\right).$$

2.16 — Question 27. Given that $x < -1$ or $x > 1$, show that

$$\operatorname{arcoth} x = \frac{1}{2}\ln\left(\frac{x+1}{x-1}\right).$$

2.16 — Question 28. Given that $x < -1$ or $x > 1$, and $k > 0$, show that

$$\operatorname{arcoth} \frac{x}{k} = \frac{1}{2} \ln \left(\frac{x+k}{x-k} \right).$$

2.16 — Question 29. Solve the equation $\operatorname{arcosech} x = \ln 3$, writing your answer in exact form.

2.16 — Question 30. Solve the equation $\operatorname{arsech} \frac{x}{4} = \ln 5$, writing your answer in exact form.

2.16 — Question 31. Solve the equation $\operatorname{arcoth} x = \ln 2$, writing your answer in exact form.

2.16 — Question 32. Solve the following equations, writing your answers in exact form:

a) $\operatorname{arcosech} x = \ln\left(2 + \frac{1}{x}\right)$

b) $\operatorname{arsech} \frac{x}{2} = \ln\left(3 - \frac{4}{x}\right)$

c) $\operatorname{arcoth} 3x = \ln\sqrt{3 + 4x}$

2.16 — Question 33. Solve the equation $\operatorname{artanh} \frac{x}{2} = \operatorname{arcoth}(3x + 1)$.

2.17 — Question 1. Write down the exponential form of each of these, giving your answer:

a) $\operatorname{cosech} x$

b) $\operatorname{sech} x$

c) $\coth x$

2.17 — Question 2. Write down the exact value of each of these in terms of exponentials, and then to 3 significant figures:

a) $\operatorname{cosech} 1$

b) $\operatorname{sech} 1$

c) $\operatorname{cosech}(-1)$

d) $\operatorname{sech}(-1)$

2.17 — Question 3. Show that $\operatorname{cosech}(-x) = -\operatorname{cosech} x$.

2.17 — Question 4. Show that $\operatorname{sech}(-x) = \operatorname{sech} x$.

2.17 — Question 5. By using the exponential forms of $\operatorname{cosech} x$, $\operatorname{sech} x$ and $\coth x$, find a simplified exact value for each of these:

a) $\operatorname{cosech}(\ln 3)$
b) $\operatorname{sech}(\ln 6)$
c) $\operatorname{cosech}(\ln \sqrt{5})$
d) $\operatorname{sech}(4 \ln 3)$

2.17 — Question 6. Show that $\operatorname{cosech}(\ln k) = \dfrac{2k}{k^2 - 1}$ where k is a positive constant.

2.17 — Question 7. Show that $\operatorname{sech}(3\ln k) = \dfrac{2k^3}{k^6+1}$ where k is a positive constant.

2.17 — Question 8. It is given that $f(x) = \operatorname{cosech}(2\ln x) - \operatorname{sech}(2\ln x)$ where $x > 0$ and $x \neq k$ for some constant k.

a) Explain why $k = 1$.

b) Show that $f(x) = \dfrac{4x^2}{(x^4-1)(x^4+1)}$.

2.17 — Question 9. By using the exponential forms of $\sinh x$ and $\cosh x$, show that $\coth x = \dfrac{e^{2x}+1}{(e^x+1)(e^x-1)}$.

2.17 — Question 10. By using the exponential forms of $\operatorname{cosech} x$, $\operatorname{sech} x$ and $\coth x$, find a simplified exact value for each of these:

a) $\coth(\ln 2)$
b) $\coth(\ln \sqrt{2})$
c) $\coth\left(\frac{1}{4}\ln 8\right)$

2.17 — Question 11. Show that $\coth(-x) = -\coth x$.

2.17 — Question 12. Find a simplified exact value for each of these:

a) $\coth(-\ln 3)$

b) $\coth(-5\ln 2)$

2.17 — Question 13. Solve the following equations, writing your answers in exact form:

a) $\operatorname{cosech} x = 1$

b) $\operatorname{cosech} x = 2$

c) $\operatorname{cosech} x = -\frac{1}{5}$

2.17 — Question 14. Solve the following equations, writing your answers in exact form:

a) $\operatorname{sech} x = 1$

b) $\operatorname{sech} x = \frac{1}{2}$

c) $\operatorname{sech} x = \frac{2}{3}$

2.17 — Question 15. Solve the following equations, writing your answers in exact form:

a) $\coth x = 2$

b) $\coth x = \frac{3}{2}$

c) $\coth x = -\frac{5}{3}$

2.17 — Question 16. Solve the following equations, writing your answers in exact form:

a) $5\operatorname{sech} x - 2\tanh x = 5$

b) $8\operatorname{cosech} x - 3\coth x = 27$

c) $2\tanh x - 5\coth x = 9$

2.17 — Question 17. Solve the following equations, writing your answers in exact form:

a) $\operatorname{cosech}^2 x = 9$

b) $\operatorname{sech}^2 x = \frac{1}{16}$

c) $\coth^2 x = 3$

2.17 — Question 18. Solve the following equations, writing your answers in exact form:

a) $\operatorname{cosech}^2 x + \operatorname{cosech} x - 20 = 0$

b) $10\operatorname{sech}^2 x - 9\operatorname{sech} x + 2 = 0$

c) $3\coth^2 x - 16\coth x - 64 = 0$

2.17 — Question 19. For each of the following functions, sketch the graph and state whether it is one-to-one or if it is many-to-one:

a) $y = \operatorname{cosech} x$
b) $y = \operatorname{sech} x$
c) $y = \coth x$

Use set notation for the following questions which require a range of values.

2.17 — Question 20. State the largest possible domain and range of the following functions:

a) $y = \operatorname{cosech} x$
b) $y = \operatorname{sech} x$
c) $y = \coth x$

2.17 — Question 21. State the range of the following functions given $x > 0$:

a) $y = \operatorname{cosech} x$

b) $y = \operatorname{sech} x$

c) $y = \coth x$

2.17 — Question 22. Sketch each of the following functions, stating their largest possible domain and range:

a) $f(x) = \operatorname{cosech}(x + 3)$

b) $g(x) = 2 + \operatorname{sech} x$

c) $h(x) = \coth(x - 1) - 1$

2.17 — Question 23. State the largest possible domain and range for each of the following functions:

a) $f(x) = \operatorname{cosech}(x-2)+2$
b) $y = 2 - \operatorname{cosech} x$
c) $g(x) = 3\operatorname{sech}(x+3)$
d) $y = 4 + 3\operatorname{sech} x$
e) $h(x) = \coth(x+4) - 3$
f) $y = 2 - \frac{1}{3}\coth x$

2.17 — Question 24. State the largest possible domain and range for each of the following functions:

a) $y = \operatorname{cosech}^2 x$
b) $y = \operatorname{sech}^2 x$
c) $y = \coth^2 x$

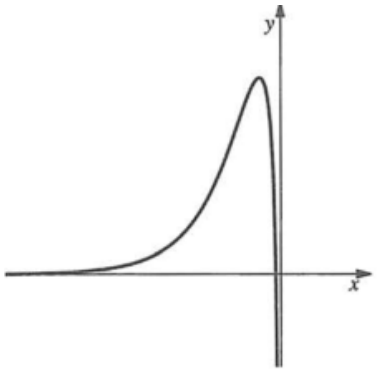
2.17 — Question 25.

a) Write $\operatorname{cosech} x + \operatorname{sech} x$ in the form $\dfrac{ae^{bx}}{e^{cx}-1}$ where a, b and c are real constants.

b) Hence explain why, for $x > 0$, $\operatorname{cosech} x + \operatorname{sech} x > \dfrac{a}{e^{cx}-1}$.

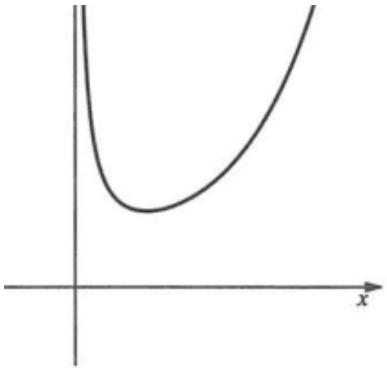
2.17 — Question 26. Show clearly that, for $x > 0$, $\coth x - \tanh x > 4e^{-2x}$.

2.17 — Question 27. The graph below shows $f(x) = \operatorname{cosech} x + 8\operatorname{sech} x$ where $x < 0$.



Find the exact largest possible range of f, writing your answer in set notation.

2.17 — Question 28. The graph below shows $f(x) = \cosh x + \coth x$ where $x > 0$.



Find the exact largest possible range of f, writing your answer in interval notation.

2.18 — Question 1. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

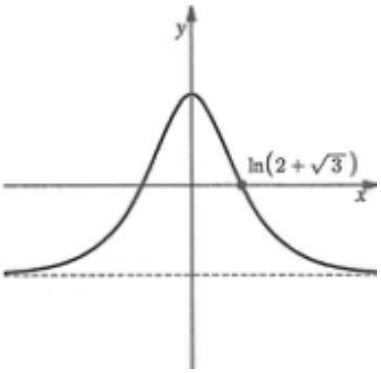
a) $y = \operatorname{cosech} x$
b) $y = \operatorname{sech} x$
c) $y = \coth x$

2.18 — Question 2.

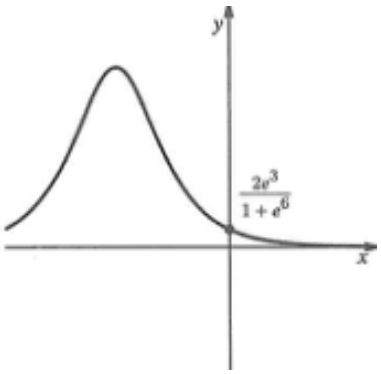
a) By sketching the curves with equations $y = \cosh x$ and $y = \operatorname{cosech} x$ on the same axes, show that they have exactly one point of intersection.
b) Find the x-coordinate of the point of intersection of the two curves, writing your answer in the form $\frac{1}{2}\ln(a+\sqrt{b})$, where a and b are positive integers.

2.18 — Question 3. Find the equation of each of the following graphs in the form $y = \operatorname{sech}(x-a)$ or $y = \operatorname{sech} x + b$, where a and b are integers:

a)



b)



2.18 — Question 4. State the equations of the asymptotes of the following graphs:

a) $y = \coth(x+1) + 1$

b) $y = \coth(x-2) - 3$

2.18 — Question 5. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \coth x + 2$
b) $y = \coth(x - 3)$
c) $y = \coth(x - 1) + 1$

2.18 — Question 6. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

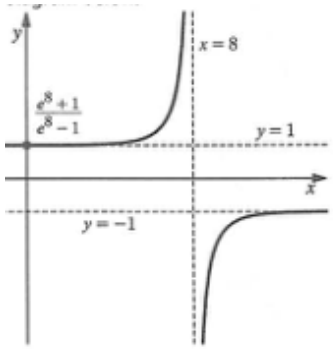
a) $y = 2\coth x$
b) $y = \frac{1}{3}\coth x$
c) $y = -\coth x$

Prerequisite knowledge for Q7–Q8: Combinations of Graph Transformations

2.18 — Question 7. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = 2 - \operatorname{cosech} x$
b) $y = -\frac{1}{3}\operatorname{sech} x$
c) $y = 2\coth x - 1$
d) $y = \operatorname{cosech}(2x + 1)$
e) $y = \operatorname{sech}(3 - x)$
f) $y = \coth\left(\frac{1}{2}x - 3\right)$

2.18 — Question 8. Suggest an equation for the curve shown in the diagram below:



2.19 — Question 1. Show, by writing $\sinh x$, $\cosh x$ and $\tanh x$ in terms of exponentials, that:

a) $\dfrac{d}{dx}(\sinh x) = \cosh x$

b) $\dfrac{d}{dx}(\cosh x) = \sinh x$

c) $\dfrac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$

Prerequisite knowledge for Q2: Hyperbolic Identities

2.19 — Question 2. It is given that $\lim_{h\to 0} \dfrac{\sinh h}{h} = 1$ and $\lim_{h\to 0} \dfrac{\cosh h - 1}{h} = 0$. Show, by first principles, that:

a) $\dfrac{d}{dx}(\sinh x) = \cosh x$

b) $\dfrac{d}{dx}(\cosh x) = \sinh x$

The following questions use the derivatives of $\sinh kx$, $\cosh kx$ and $\tanh kx$ and require no further Chain, Product or Quotient Rule.

2.19 — Question 3. Differentiate each of the following:

a) $y = \sinh 2x$
b) $y = \cosh 3x$
c) $y = \tanh 6x$
d) $y = \sinh \frac{1}{4}x$
e) $y = \cosh \frac{x}{3}$
f) $y = \tanh\left(-\frac{x}{5}\right)$

2.19 — Question 4. Differentiate each of the following:

a) $y = 3\sinh 4x - 6\tanh 3x$
b) $y = 5\cosh \frac{x}{10} + \frac{1}{2}\tanh \frac{x}{4}$

2.19 — Question 5. Find the exact gradient of the curve $y = 3\tanh\frac{1}{2}x$ at $\left(\ln 16, \frac{45}{17}\right)$.

2.19 — Question 6. Find the equation of the tangent to the curve $y = 3\sinh x$ when $x = \ln 6$, writing your answer in the form $ax + by = c$ where a, b and c are constants.

2.19 — Question 7. Find the equation of the tangent to the curve $y = 5\sinh x - \cosh x$ when $x = \ln 2$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

2.19 — Question 8. Find the equation of the tangent to the curve $y = 3\cosh 2x - 4\sinh 3x$ when $x = \ln \frac{1}{3}$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

2.19 — Question 9. Find the equation of the tangent to the curve $y = \tanh \frac{x}{2}$ when $x = \ln 4$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

2.19 — Question 10. Find the equation of the normal to the curve $y = 8 + 6\tanh \frac{1}{3}x - 4\tanh \frac{2}{3}x$ when $x = 0$, writing your answer in the form $y = ax + b$ where a and b are constants in exact form.

2.19 — Question 11. The line l_1 is the normal to the curve $y = 12\sinh\frac{x}{3}$ when $x = \ln 8$, and the line l_2 is the normal to the curve $y = 6\cosh\frac{x}{4}$ when $x = \ln 16$. Show that the lines l_1 and l_2 intersect when $x = a\ln 2 + b$, where a and b are rational numbers to be found.

2.19 — Question 12. Show that the curve with equation $y = \sinh x + \tanh x$ is always increasing.

2.19 — Question 13. It is given that the curve with equation $y = \sinh x + 2\cosh x$ has a single stationary point. Find the exact coordinates of the stationary point. **Prerequisite knowledge for Q14: Hyperbolic Double-Angle Formulae**

2.19 — Question 14. Show that the curve with equation $y = 9\cosh x + 3\sinh 2x$ has no point where the gradient is 1.

2.19 — Question 15. It is given that $f(x) = p\sinh x - q\cosh x$ where p and q are positive constants. Show that the curve $y = f(x)$ has no stationary points when $p \geq q$.

2.19 — Question 16. It is given that the curve with equation $y = \sinh x - 8\tanh x$ has two stationary points. Find the exact coordinates of both stationary points.

2.19 — Question 17. Show that the curve $y = 5\cosh x + \sinh x + x$ has a single stationary point.

2.19 — Question 18. Determine whether the curve with equation $y = \cosh 3x - 2\sinh x$ is convex (concave upwards) or concave (concave downwards) when $x = \ln 2$.

2.19 — Question 19. Show that the curve $y = 4\sinh 2x - 3\cosh 2x$ has a single point of inflection and find its exact coordinates. **The following questions use the derivatives of** $\sinh kx$**,** $\cosh kx$ **and** $\tanh kx$ **and use any differentiation knowledge from A-Level Maths.**

2.19 — Question 20. Differentiate each of the following:

a) $y = \sinh(\sqrt{x})$

b) $y = \cosh\left(\frac{1}{\sqrt{x}}\right)$

c) $y = \tanh\left(\frac{x+1}{2\sqrt{x}}\right)$

2.19 — Question 21. Differentiate each of the following:

a) $y = \sinh^2 x$
b) $y = \cosh^2 x$
c) $y = \sinh^3 x$
d) $y = \cosh^4 x$
e) $y = \tanh^2 x$
f) $y = \tanh^5 x$

2.19 — Question 22. Show that the curve $y = 3\cosh^2 x - \sinh x$ has a single stationary point.

2.19 — Question 23. It is given that $f(x) = \cosh^2 2x - p\cosh 2x$, where p is a constant. Find the minimum value of p such that $y = f(x)$ has more than one stationary point.

2.19 — Question 24. Differentiate each of the following:

a) $y = \sinh(\cosh x)$

b) $y = \cosh(\sinh x)$

c) $y = \tanh(\tanh x)$

2.19 — Question 25. Show that:

a) $\dfrac{d}{dx}(\ln(\sinh x)) = \coth x$

b) $\dfrac{d}{dx}(\ln(\cosh x)) = \tanh x$

Prerequisite knowledge for Q26: Hyperbolic Double-Angle Formulae

2.19 — Question 26. Show that $\dfrac{d}{dx}\left(\dfrac{1}{2}\ln(\tanh x)\right) = \operatorname{cosech} 2x$.

2.19 — Question 27. Differentiate each of the following:

a) $y = \sinh x \cosh x$
b) $y = \sinh 3x \cosh 4x$
c) $y = x \sinh x$
d) $y = x^2 \cosh x$
e) $y = (x+1)\sinh 2x$
f) $y = (1-x^2)\cosh \frac{x}{2}$

2.19 — Question 28. Find the exact gradient of the curve $y = \sinh 2x \tanh 3x$ at $\left(\ln 2, \frac{189}{104}\right)$.

2.19 — Question 29. Find the equation of the tangent to the curve $y = (x^2 + x - 6)\cosh x$ when $x = 0$, writing your answer in the form $y = ax + b$ where a and b are integers.

2.19 — Question 30. Find the equation of the normal to the curve $y = \sinh(12x - \ln 2)\cosh(3x - \ln 3)$ when $x = 0$, writing your answer in the form $ax + by = c$ where a, b and c are integers.

2.19 — Question 31. Show that the curve with equation $y = \sinh x \cosh x$ is always increasing.

2.19 — Question 32. It is given that the curve with equation $y = (\cosh x - 3)\cosh x$ has three stationary points. Find the exact coordinates of all three stationary points.

2.19 — Question 33. Determine whether the curve with equation $y = \sinh\frac{x}{2}\left(2 - \sinh\frac{x}{3}\right)$ is convex (concave upwards) or concave (concave downwards) when $x = \ln 64$.

2.19 — Question 34. Show that the curve $y = (\sinh x - 6)\sinh x$ has two points of inflection and find their exact coordinates.

2.19 — Question 35. By writing $\tanh x$ as $\dfrac{\sinh x}{\cosh x}$, use the quotient rule to show that $\dfrac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$.

2.19 — Question 36. Differentiate each of the following:

a) $y = \dfrac{\sinh x}{\cosh 2x}$

b) $y = \dfrac{\cosh x}{\sinh 2x}$

c) $y = \dfrac{x}{\sinh x}$

d) $y = \dfrac{\cosh x}{x}$

e) $y = \dfrac{x+1}{\sinh 2x}$

f) $y = \dfrac{\cosh 3x}{x^2+1}$

2.19 — Question 37. Find the exact gradient of the curve $y = \dfrac{\sinh 3x}{\sinh x}$ at $\left(\ln 2, \frac{21}{4}\right)$.

2.19 — Question 38. Find the equation of the tangent to the curve $y = \dfrac{x + \cosh 3x}{1 - \sinh 2x}$ when $x = 0$, writing your answer in the form $y = ax + b$ where a and b are integers.

2.19 — Question 39. Find the equation of the normal to the curve $y = \dfrac{\tanh 3x + 6}{\cosh 2x + 3}$ when $x = 0$, writing your answer in the form $ax + by = c$ where a, b and c are integers.

2.19 — Question 40. It is given that $f(x) = \dfrac{x+1}{\sinh x}$.

a) Show that the x-coordinates of the stationary points of the curve $y = f(x)$ satisfy $\tanh x = x + 1$.

b) By sketching $y = \tanh x$ and $y = x + 1$ on the same axes, determine the number of stationary points the curve $y = f(x)$ has.

2.19 — Question 41. It is given that the curve with equation $y = \dfrac{\sinh x + 1}{\tanh x}$ has a single stationary point. Find the exact coordinates of the stationary point.

2.19 — Question 42. It is given that the curve with equation $y = \dfrac{\cosh x}{4 - \sinh x}$ has a gradient of $\frac{1}{16}$ at two points. One of these points is $x = 0$. Find the exact x-coordinate of the other point.

2.19 — Question 43. Show that the curve $y = \dfrac{4}{\cosh x}$ has two points of inflection and find their exact coordinates.
The following questions use the derivatives of $\operatorname{sech} x$, $\operatorname{cosech} x$ **and** $\coth x$ **and use any differentiation knowledge from A-Level Maths.**

2.19 — Question 44. By writing $\coth x$ as $\dfrac{\cosh x}{\sinh x}$, use the quotient rule to show that $\dfrac{d}{dx}(\coth x) = -\operatorname{cosech}^2 x$.

2.19 — Question 45. By writing $\operatorname{cosech} x$ as $\dfrac{1}{\sinh x}$, use the quotient rule to show that

$$\frac{d}{dx}(\operatorname{cosech} x) = -\coth x \operatorname{cosech} x.$$

2.19 — Question 46. By writing $\operatorname{sech} x$ as $\dfrac{1}{\cosh x}$, use the quotient rule to show that $\dfrac{d}{dx}(\operatorname{sech} x) = -\tanh x \operatorname{sech} x$.

2.19 — Question 47. Show that the curve with equation $y = 2x + \operatorname{sech} x$ has two points of inflection. **Prerequisite knowledge for Q48: Hyperbolic Identities**

2.19 — Question 48. Show that $\dfrac{d}{dx}\left(\ln \tanh \dfrac{x}{2}\right) = \operatorname{cosech} x$. **Prerequisite knowledge for Q49: Differentiation of Inverse Trigonometric Functions**

2.19 — Question 49. Given that $y = \arctan(4e^{5x})$, show that $\dfrac{dy}{dx} = \dfrac{20}{p\sinh 5x + q\cosh 5x}$ where p and q are integers to be found.

2.20 — Question 1. Find:

a) $\displaystyle\int \sinh x\,dx$

b) $\displaystyle\int \cosh x\,dx$

c) $\displaystyle\int \operatorname{sech}^2 x\,dx$

d) $\displaystyle\int \sinh 3x\,dx$

e) $\displaystyle\int \cosh \frac{x}{2}\,dx$

f) $\displaystyle\int \operatorname{sech}^2 9x\,dx$

2.20 — Question 2. Evaluate the following definite integral:

$$\int_{\ln 2}^{\ln 3} \sinh x \, dx$$

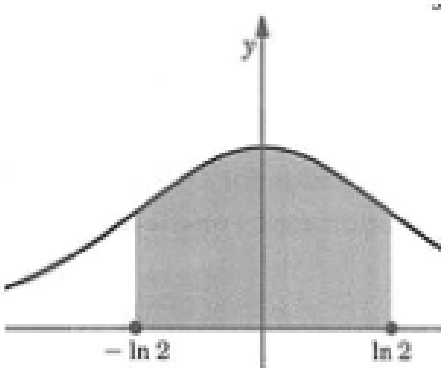
2.20 — Question 3. Evaluate the following definite integral:

$$\int_{\ln 3}^{\ln 4} \cosh 2x \, dx$$

2.20 — Question 4. Evaluate the following definite integral:

$$\int_0^{\ln 2} (15\cosh 3x - 9\sinh 3x)\,dx$$

2.20 — Question 5. The diagram below shows the curve $y = \operatorname{sech}^2 x$. Show that the exact shaded area is $\frac{6}{5}$.



2.20 — Question 6. Find the exact value of:

$$\lim_{\delta x \to 0} \sum_{x=0}^{\ln 2} (8\cosh 4x - 5\sinh 3x)\,\delta x$$

2.20 — Question 7. It is given that $f(x) = 6\sinh 3x + 15\cosh 3x$. Find the exact positive value of k for which the region between the curve $y = f(x)$, the x-axis, the y-axis, and the line $x = k$ is 8. **Prerequisite knowledge for Q8: Integration of Reciprocal Trigonometric Functions**

2.20 — Question 8. Show that

$$\int_{-\pi/4}^{\pi/4} (\sec^2 x - \operatorname{sech}^2 x)\, dx > \frac{4}{1+e^{\pi}}.$$

2.20 — Question 9. Find:

a) $\displaystyle\int \sinh(2x+5)\, dx$

b) $\displaystyle\int 8\cosh(3-4x)\, dx$

c) $\displaystyle\int 2x \operatorname{sech}^2 x^2\, dx$

d) $\displaystyle\int x\sinh(3x^2-2)\, dx$

e) $\displaystyle\int \frac{\cosh\sqrt{x}}{2\sqrt{x}}\, dx$

f) $\displaystyle\int \frac{\operatorname{sech}^2(3/x^2)}{4x^3}\, dx$

2.20 — Question 10. Evaluate the following definite integral:

$$\int_0^{\ln 2} \sinh\left(\frac{3x + \ln 4}{2}\right) dx$$

2.20 — Question 11. Evaluate the following definite integral:

$$\int_0^{\sqrt{\ln 2}} 256x \cosh(4x^2)\, dx$$

2.20 — Question 12. Show that:

$$\int_0^{\pi/2} \cos x \sinh(\sin x)\, dx = \frac{(e-1)^2}{2e}.$$

2.20 — Question 13. Find:

a) $\int \tanh x\, dx$

b) $\int \coth x\, dx$

2.20 — Question 14. Show that

$$\int_{\ln 2}^{\ln 4} (2\tanh 2x + 2\coth 2x)\,dx = \ln\left(\frac{p}{q}\right),$$

where p and q are integers and $\frac{p}{q}$ is in its simplest form.

2.20 — Question 15.

a) Find $\dfrac{d}{dx}(\operatorname{cosech} x)$.

b) Hence find $\displaystyle\int \operatorname{cosech} x \coth x\,dx$.

2.20 — Question 16.

a) Find $\frac{d}{dx}(\operatorname{sech} x)$.

b) Hence find $\int \operatorname{sech} x \tanh x \, dx$.

2.20 — Question 17.

a) Find $\frac{d}{dx}(\coth x)$.

b) Hence find $\int \operatorname{cosech}^2 x \, dx$.

The following questions in this section require the use of Integration by Substitution, Integration by Parts, Hyperbolic Double Angle Formulae and Partial Fractions.

2.20 — Question 18. Find:

a) $\displaystyle\int x \sinh 3x \, dx$

b) $\displaystyle\int x^2 \cosh \frac{x}{4} \, dx$

2.20 — Question 19. The curve has gradient function

$$\frac{dy}{dx} = 9(x+2)\sinh 3x$$

and passes through the point $(0, 8)$. Show that y can be written in the form

$$y = (ax+b)\cosh 3x - \sinh 3x + c,$$

where a, b and c are integers.

2.20 — Question 20. Find:

a) $\int e^{x} \sinh x \, dx$

b) $\int e^{2x} \cosh x \, dx$

c) $\int e^{3x} \sinh 2x \, dx$

d) $\int e^{-4x} \cosh 4x \, dx$

2.20 — Question 21. Find $\int \cosh^{2} x \, dx$.

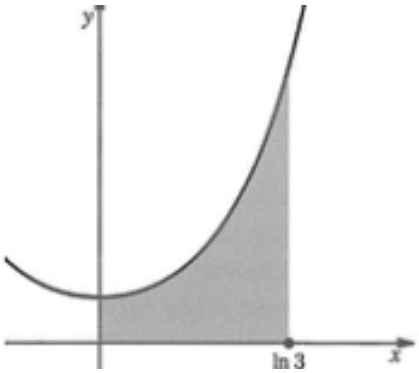
2.20 — Question 22. Find $\int \sinh^2 6x \, dx$.

2.20 — Question 23. Show that

$$\int_{\ln(5/6)}^{\ln(6/5)} \cosh^2 \frac{x}{2} \, dx = p + \ln q,$$

where p and q are positive rational numbers.

2.20 — Question 24. The diagram below shows the curve $y = 4\cosh^2 x - 3\cosh x$. Show that the exact shaded area is $\frac{4}{9} + \ln 9$.



2.20 — Question 25. Find:

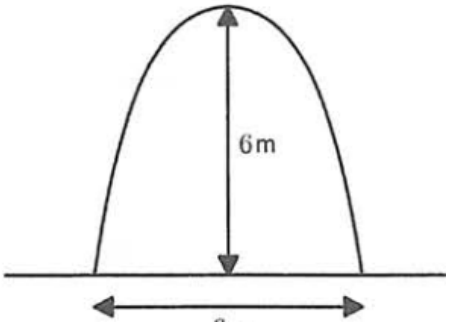
a) $\int \sinh^3 x \, dx$

b) $\int \cosh^3 x \, dx$

2.21 — Question 1. The cross-section of a tunnel is modelled by the curve with equation

$$y = 2p - q\cosh x$$

where y is the height, in metres, of the tunnel above the road surface, x is the horizontal distance, in metres, from the centre of the tunnel, and p and q are positive constants. The tunnel is symmetric about the vertical line $x = 0$.



The tunnel is 6 metres wide at road level and has a maximum height of 6 metres.

a) Find the values of p and q to three significant figures.

b) A vehicle of rectangular cross-section is to pass through the tunnel. The vehicle is 4 metres wide and 4 metres high. Determine whether the vehicle can pass through the tunnel without touching the sides.

Prerequisite knowledge for Q2: Mean Value and Reversing the Chain Rule

2.21 — Question 2. The temperature, $T°\text{C}$, of a chemical solution t hours after a catalyst is added can be modelled by the equation

$$T = 60 - 15\tanh 0.8t$$

for $t \geq 0$. Using the model, determine the mean temperature of the solution during the first 45 minutes of the catalyst being added, giving your answer correct to two decimal places.

2.21 — Question 3. The velocity, $v\,\mathrm{m\,s^{-1}}$, of a skydiver t seconds after leaving a stationary helicopter is modelled by

$$v = 55\tanh(0.12t)$$

for $t \geq 0$.

a) Explain why this model predicts the existence of a terminal velocity, and state its value.

b) According to the model, find the time taken for the skydiver to reach 90% of the terminal velocity. Give your answer to 2 decimal places.

Prerequisite knowledge for Q4: Volume of Revolution and Hyperbolic Identities

2.21 — Question 4. A water treatment company is designing a large filtration pipe. The inside wall of the pipe is modelled by the curve

$$y = 2 + \cosh\left(\frac{x}{2}\right)$$

for $-4 \leq x \leq 4$, where x and y are measured in metres. The pipe is formed by rotating the region enclosed by the curve and the x-axis about the x-axis.

a) Show that the volume V of the pipe is 381 m^3 to the nearest cubic metre.

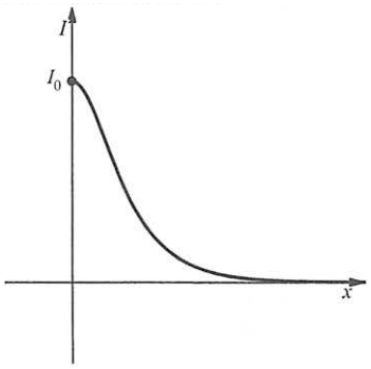
b) The pipe can process 0.85 m^3 of water per second when completely full. Estimate how many seconds are needed to completely fill the pipe.

Prerequisite knowledge for Q5: Differentiation of Hyperbolic Functions

2.21 — Question 5. Light travelling through water decreases in intensity with depth. In a particular model, the intensity I at a depth of x metres below the surface is given by

$$I = I_0 \operatorname{sech}(kx)$$

where I_0 is the intensity at the surface, k is a positive constant, and $x \geq 0$.



a) Show that the depth x at which the intensity is equal to fI_0, where $0 < f < 1$, is given by $x = \dfrac{1}{k}\operatorname{arcosh}\left(\dfrac{1}{f}\right)$.

b) Given that the intensity is reduced to 40% of its surface value at a depth of 12 metres, find the value of k, giving your answer to 3 significant figures.

c) Given that $I_0 = 120$, find an expression for $\dfrac{dI}{dx}$ in terms of k and x.

d) Using your value of k from part (b), find the rate of change of light intensity with respect to depth when $x = 8$, giving your answer to 3 significant figures.

e) Show that the magnitude of $\dfrac{dI}{dx}$ decreases as x increases for $x \geq 0$. Hence explain what this suggests about the behaviour of light as it enters the water.

f) Given that $\operatorname{sech} u \approx 1 - \frac{u^2}{2}$ for small values of u, comment on how well the model represents light intensity close to the surface.

2.21 — Question 6.

a) Show that $e^{kt} \equiv \cosh kt + \sinh kt$.

The displacement, x metres, of a particle at time t seconds is governed by the differential equation

$$\frac{d^2x}{dt^2} - 9x = 0.$$

Initially, the particle is at $x = 2$ and is moving with velocity 3 $\mathrm{m\,s^{-1}}$.

b) Show that the general solution can be written in the form $x = A\cosh 3t + B\sinh 3t$, where A and B are constants to be found.

c) Express the solution in the form $x = R\cosh(3t + \alpha)$.

d) Find the time at which the displacement of the particle is $2\sqrt{3}$ metres, giving your answer in the form $\dfrac{1}{6}\ln(a + b\sqrt{3})$, where a and b are rational numbers.

Prerequisite knowledge for Q7: Differentiation and Integration of Hyperbolic Functions

2.21 — Question 7. A small research rocket is launched vertically upwards. Its velocity, $v\,\mathrm{m\,s^{-1}}$, at time t seconds after launch is modelled by the equation

$$v = A\sinh kt$$

where A and k are positive constants.

a) Show that the acceleration of the rocket, $a\,\mathrm{m\,s^{-2}}$, can be expressed in the form $a = k\sqrt{v^2 + A^2}$.

b) It is given that the initial acceleration of the rocket is $2\ \mathrm{m\,s^{-2}}$. It is also known that when the rocket reaches a velocity of $12\ \mathrm{m\,s^{-1}}$, its acceleration is $5.2\ \mathrm{m\,s^{-2}}$. Find the value of A.

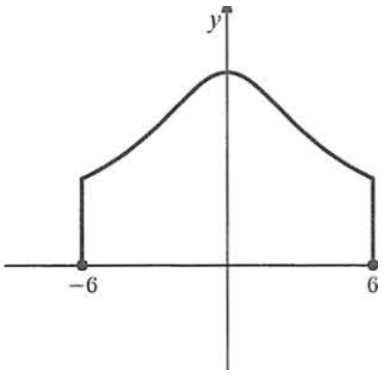
c) Show that $k = 0.4$.

d) Find the distance travelled by the rocket in the first 5 seconds of its launch, giving your answer to the nearest metre.

Prerequisite knowledge for Q8: Integration using Inverse Hyperbolic Functions

2.21 — Question 8. A farmer is designing a new barn for storing hay. The cross-section of the barn is modelled for $-6 \le x \le 6$ by the curve

$$y = \frac{12}{\sqrt{x^2 + 9}}.$$



The region enclosed by the curve, the x-axis, and the lines $x = -6$ and $x = 6$ represents the cross-sectional area of the barn. Each unit on the axes represents 2 metres. The barn is modelled as a prism of length 18 metres.

a) Show that the cross-sectional area of the barn is $96\ln(2 + \sqrt{5})\,\mathrm{m^2}$.

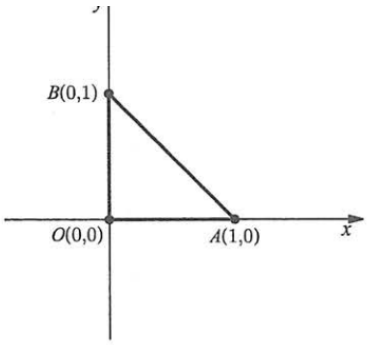
b) Find the volume of the barn, giving your answer to the nearest $\mathrm{m^3}$.

c) The farmer stores hay bales occupying $1.8\ \mathrm{m^3}$. Estimate the maximum number of hay bales that can fit inside the barn.

2.22 — Question 1. The transformation T is represented by the matrix $M = \begin{pmatrix} \cosh x & \sinh x \\ \sinh x & \cosh x \end{pmatrix}$, where $x > 0$.

a) Show that T preserves area.

b) Find the exact value of x that maps the triangle shown in the diagram below onto an equilateral triangle.



Prerequisite knowledge for Q2: Volume of Revolution, Hyperbolic Identities

2.22 — Question 2. It is given that $f(x) = 4\sinh x - 6\cosh x$.

a) Show that the curve $y = f(x)$ has a turning point at $x = \frac{1}{2}\ln 5$ and determine its type.

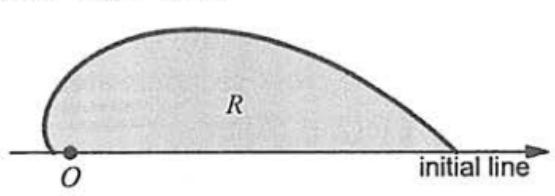
b) The arc of the curve $y = f(x)$ between $x = 0$ and $x = \frac{1}{2}\ln 5$ is rotated through 2π radians about the x-axis. Show that V, the volume generated, is given by $\pi(p + q\ln 5)$, where p and q are integers.

Prerequisite knowledge for Q3: Polar Integration

2.22 — Question 3. The diagram below shows the curve with polar equation

$$r = 4\sqrt{\cosh 2\theta - \sinh 2\theta},$$

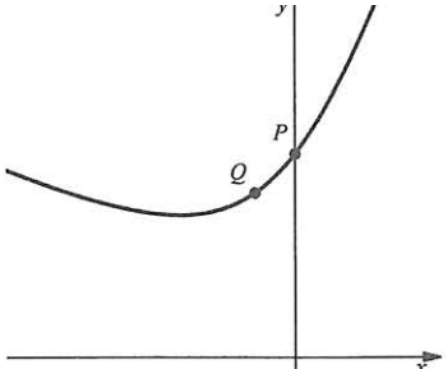
where $0 \leq \theta \leq 2\pi$.



The region R is bounded by the initial line, the curve, and the line with $\theta = \pi$. Find the exact area of R, giving your answer in the form $p + qe^{r\pi}$, where p, q and r are integers. **Prerequisite knowledge for Q4–Q5: Parametric Equations and Hyperbolic Differentiation**

2.22 — Question 4. The diagram below shows the curve with parametric equations

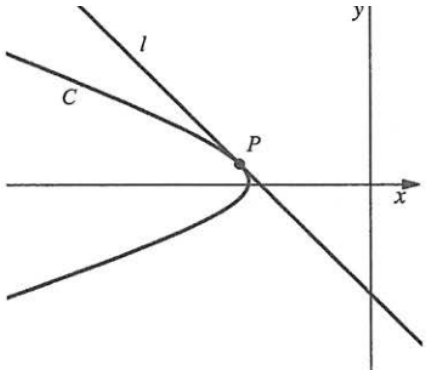
$$x = 3\sinh t - \cosh t \qquad \text{and} \qquad y = \sinh t + 2\cosh t.$$



a) The curve intersects the y-axis at P. Find the exact coordinates of the point P.
b) Q is the point on the curve when $t = 0$. Find the equation of the tangent to the curve at Q in the form $ax+by = c$, where a, b and c are integers.

2.22 — Question 5. The diagram below shows the curve C with parametric equations

$$x = 5\sinh t - 8\cosh t \qquad \text{and} \qquad y = 2\cosh t - 3\sinh t.$$

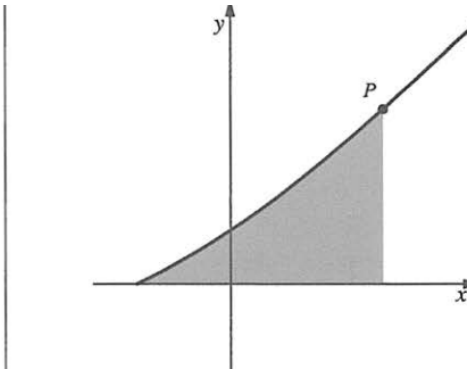


The line l is a tangent to the curve at P, where $x + y$ is at its maximum. Find the equation of the line l in the form $x + y = k$, where k is an exact constant. **Prerequisite knowledge for Q6: Parametric Equations and Hyperbolic Identities**

2.22 — Question 6. The diagram below shows the curve with parametric equations

$$x = 2\sinh 2t - \cosh 2t, \qquad y = \sinh 2t$$

for $t \geq 0$. It is given that P is the point on the curve with $t = \ln 2$. The shaded region is the region enclosed by the curve, the x-axis, and the vertical line from the x-axis through P.



a) Show that the Cartesian equation of the curve is $x^2 + 3y^2 = 4xy + 1$.
b) Show that the area of the shaded region can be written in the form $a + \ln b$, where a is a rational number in its simplest form and b is a positive integer.

Prerequisite knowledge for Q7: Euler's Formula

2.22 — Question 7. It is given that z is a complex number.

a) By using Euler's formula, show that $\cosh(iz) = \cos z$.

b) Assuming that the equation has complex roots, find in logarithmic form the roots of the equation $\cos z = 4$.

Prerequisite knowledge for Q8: Integration with Inverse Trigonometric Functions

2.22 — Question 8.

a) Show that $3\sinh x + 5\cosh x$ can be written in the form $e^{-x}(p + qe^{2x})$, where p and q are integers.

b) Hence, using a suitable substitution, show that

$$\int_{\ln(\sqrt{3}/6)}^{\ln(1/2)} \frac{1}{3\sinh x + 5\cosh x}\, dx = \frac{\pi}{24}.$$

Prerequisite knowledge for Q9: Maclaurin Series

2.22 — Question 9. It is given that

$$f(x) = \operatorname{artanh}\left(\frac{2x-1}{2-2x}\right), \qquad |x| < \frac{3}{4}.$$

a) Write $f(0)$ in the form $\ln\left(\frac{\sqrt{k}}{k}\right)$, where k is a positive integer.

b) Show that $f'(x) = \frac{2}{3-4x}$.

c) Find $f''(x)$.

d) Hence show that the Maclaurin series for $f(x)$, up to and including the x^2 term, can be written as $\ln\left(\frac{\sqrt{k}}{k}\right) + px + qx^2$, where p and q are constants to be found.

Prerequisite knowledge for Q10: De Moivre's Theorem and Mean Value

2.22 — Question 10. It is given that $f(x) = \sinh ix$, where $0 \leq x \leq 2\pi$. Find the mean value of $y = (f(x))^4$.

2.22 — Question 11.

a) Given that $\cosh(A+B) = \cosh A \cosh B + \sinh A \sinh B$, express $\cosh((p+1)x)$ and $\cosh((p-1)x)$ in terms of $\cosh(px)$, $\sinh(px)$, $\cosh x$, and $\sinh x$.

b) Hence show that

$$\sinh(px) = \frac{\cosh((p+1)x) - \cosh((p-1)x)}{2\sinh x}.$$

c) Hence find the sum of the series $S_n = \sinh x + \sinh 2x + \cdots + \sinh nx$ in terms of $\cosh x$, $\sinh x$, $\cosh nx$, and $\cosh((n+1)x)$.

Prerequisite knowledge for Q12–Q13: Integrating Factors for First Order Differential Equations

2.22 — Question 12. Given that $y = 2$ when $x = \ln 2$, solve the differential equation

$$\frac{dy}{dx} + y\coth x = \cosh^2 x.$$

2.22 — Question 13. Given that $y = 5$ when $x = 0$, solve the differential equation

$$\frac{dy}{dx} - y \tanh x = \sinh 2x.$$